CAXTON OJIAMBO

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PROFESSIONAL SUMMARY

Results-oriented Senior Systems Engineer specializing in Model-Based Systems Engineering (MBSE), software architecture, and cybersecurity integration for mission-critical defense and aerospace systems. Skilled at applying MBSE frameworks (SysML/UML) to design, analyze, and validate complex systems throughout their lifecycle. Experienced in developing secure, scalable software aligned with DoD standards (RMF, STIG, DoDI 5000, NIST 800-53). Recognized for leading cross-functional teams to deliver enterprise-grade, secure, and high-performance solutions.

CORE COMPETENCIES

Model-Based Systems Engineering (MBSE) | Systems Architecture | SysML / UML | DevSecOps | Agile Development | CI/CD | Software Engineering | Cybersecurity (RMF, STIG, NIST 800-53) | AI/ML Integration | Cloud-Native Systems | Docker | Kubernetes | Systems Integration | DoDAF | Requirements Analysis | Verification & Validation | Technical Leadership | Documentation & Compliance

PROFESSIONAL EXPERIENCE

Senior Systems Engineer (MBSE) – Department of Defense | June 2024 – Present

- Translated complex system requirements into formal MBSE models ensuring traceability and validation across the lifecycle.
- Decomposed system architectures into well-defined subsystem components and interfaces to enable seamless integration.
- Estimated technical resources and supported program management in planning, cost analysis, and scheduling.
- Developed and managed system models in Cameo, MagicDraw, Rhapsody, or Enterprise Architect to document architecture, requirements, and behaviors.
- Created SysML/UML diagrams including activity, sequence, and block definition diagrams (BDDs, IBDs).
- Applied MBSE to support system-of-systems design, trade studies, and risk reduction.
- Integrated MBSE models with tools such as DOORS, MATLAB/Simulink, and simulation frameworks.
- Led system architecture definition, interface management, and baseline control for defense programs.
- Ensured alignment with DoDAF standards, RMF cybersecurity requirements, and MIL-STDs.
- Collaborated across software, hardware, and cybersecurity teams to deliver cohesive system solutions.
- Supported acquisition, testing, and sustainment activities with digital engineering deliverables.

Senior System/Software Engineer – Boeing | Aug 2022 – June 2024

- Designed, developed, and integrated mission-critical software for aerospace and defense systems.
- Led and executed system integration and troubleshooting efforts, collaborating across electrical, software, and mechanical teams to resolve technical issues rapidly.
- Developed test frameworks and automation scripts in Python to streamline verification and data analysis processes.
- Produced comprehensive system design documentation, user manuals, and customer support

materials.
- Collaborated closely with program managers, SMEs, and customers to ensure delivery of robust, mission-ready solutions.
- Supported domestic and international field testing and validation activities, including hands-on radar or RF system deployment.
- Defined and documented system requirements by analyzing customer applications, mission goals, and operational constraints.
- Provided technical leadership across full lifecycle phases — design, development, procurement, integration, and validation of system solutions.
- Applied systems engineering to define architecture, interfaces, and verification plans.
- Programmed in C++, C#, Java, Python, and Ada for embedded and real-time applications.
- Led design reviews (SRR, PDR, CDR) and participated in performance trade-offs and risk analysis.
- Served as SME in system/software architecture and integration.
- Managed requirements traceability between system-level design, implementation, and testing.
- Drove adoption of Agile, DevSecOps, and MBSE practices to improve efficiency and quality.

System/Software Engineer – DoD / National Security Agency | Apr 2021 – Aug 2022

- Collaborated with systems engineers to integrate software with embedded, networked, and enterprise systems.
- Conducted vulnerability assessments and implemented security features in compliance with DoD and NSA cybersecurity standards (RMF, STIGs).
- Participated in system-level design reviews and contributed to architecture decisions for mission systems.
- Maintained and enhanced legacy systems while planning and implementing upgrades or replacements.
- Developed and executed automated test plans and supported verification/validation activities in secure environments.
- Prepared technical documentation, including design specs, interface documents, and configuration management records.
- Provided engineering support during deployment and sustainment phases of the system lifecycle.
- Supported classified operations under DoD 8570/8140 cybersecurity workforce compliance (if applicable).
- Worked within Agile or DevSecOps teams to deliver iterative, tested, and deployable software in secure environments.
- Integrated and tested hardware subsystems, including radar and communications components, within embedded and networked environments.
- Applied strong troubleshooting and debugging skills across hardware, software, and system interfaces to identify and resolve root causes.
- Ensured system security and compliance with DoD or government standards, including operating system hardening and network integrity.
- Designed and developed scalable, secure software solutions aligned with DoD mission requirements.
- Built real-time, mission-critical software for C2, ISR, and defense support systems.
- Conducted field tests and deployments for operational readiness validation.
- Authored software design descriptions (SDD), interface documents (ICD), and test plans.

Cyber / Systems Software Engineer – Northrop Grumman | Apr 2020 – Apr 2021

- Designed secure, mission-grade software architectures under DoD cybersecurity frameworks.
- Conducted threat modeling, vulnerability analysis, and risk assessments.
- Utilized Linux environments (Ubuntu, RedHat, Yocto) for system integration, testing, and performance verification.

- Employed data analysis and visualization techniques to assess system performance and validate design effectiveness.
- Integrated cyber defense measures (intrusion detection, anomaly monitoring) within embedded systems.
- Collaborated with systems and test teams for V&V; and reliability analysis.
- Produced comprehensive documentation including SRS, SDD, and ICD deliverables.

Cybersecurity Engineer / Analyst – AmTote International Inc. | Mar 2018 – Apr 2020

- Analyzed alerts and incidents across IDS/IPS, antivirus, EDR, and DLP systems.
- Conducted triage, containment, and remediation for enterprise security incidents.
- Tuned SIEM and detection systems to enhance accuracy and reduce false positives.
- Reported security metrics, incident trends, and compliance status to leadership.
- Monitored patch and vulnerability management using Nessus and Qualys.

System/Software Application Developer – Hood College | Apr 2017 – Mar 2018

- Designed, developed, and maintained enterprise systems supporting academic and administrative functions.
- Developed scalable, maintainable code in languages such as C++, Java, Python, or embedded C for real-time and enterprise systems.
- Integrated software components with existing systems, ensuring interoperability across classified and unclassified networks.
- Collaborated with cross-functional teams including cybersecurity, systems engineering, and intelligence analysts.
- Programmed in Java, Python, PHP, JavaScript, and SQL to automate campus workflows.
- Customized SIS and LMS systems and built mobile-friendly web applications.
- Created optimized database queries and generated institutional reports for multiple departments.

Software Engineer – Beyond Care | Jun 2014 – Apr 2017

- Developed secure healthcare and telemedicine applications, including EHRs and patient portals.
- Built APIs and backend integrations for healthcare databases and external systems.
- Ensured HIPAA compliance with encryption, RBAC, and secure data handling practices.
- Conducted code and security reviews to maintain regulatory and performance standards.

EDUCATION

Doctor of Science – Information Technology, Towson University

M.S. – Computer Science (Data Science), Towson University, 2020

B.S. – Computer Science, University of Maryland Baltimore County / Towson University, 2017